

CMPS 03

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Class Discussion

Why are *differential equations* an important technique for computational modelling in psychology and neuroscience?

1 Dynamics

Spiking neuron models rely on differential equation to capture the evolution of voltage as a function of time. Dynamics is the study of things changing over time, and neural dynamics gives us important insights into brain and neuronal activity normally and as a consequence of disease.

An interest in dynamical systems has been a part of computational neuroscience since the days of Hodgkin and Huxley (more on them later). But with the development of cheaper and more powerful computing capacities their use as expanded practically and theoretically. It is increasingly feasible to simulate complex models in high dimensions. This advance parallels the use of multiple electrode recordings that yield high dimensional data.

To ground our understanding with the use of differential equations in computational neuroscience we will produce simple, single neuron simulations of the integrate and fire variety. An excellent and concise summary of these ideas, one that I drew upon heavily for this course is.¹

1.1 Beginning to Think Dynamically

Assume you have a derivative that is equal to $x - x^3$ (see Figure 1). What would it mean to "solve" this equation?

¹David Terman, "An Introduction to Dynamical Systems and Neuronal Dynamics," in *Tutorials in Mathematical Biosciences I* (Springer Berlin Heidelberg, 2005), 21–68, https://doi.org/10.1007/978-3-540-31544-5_2.

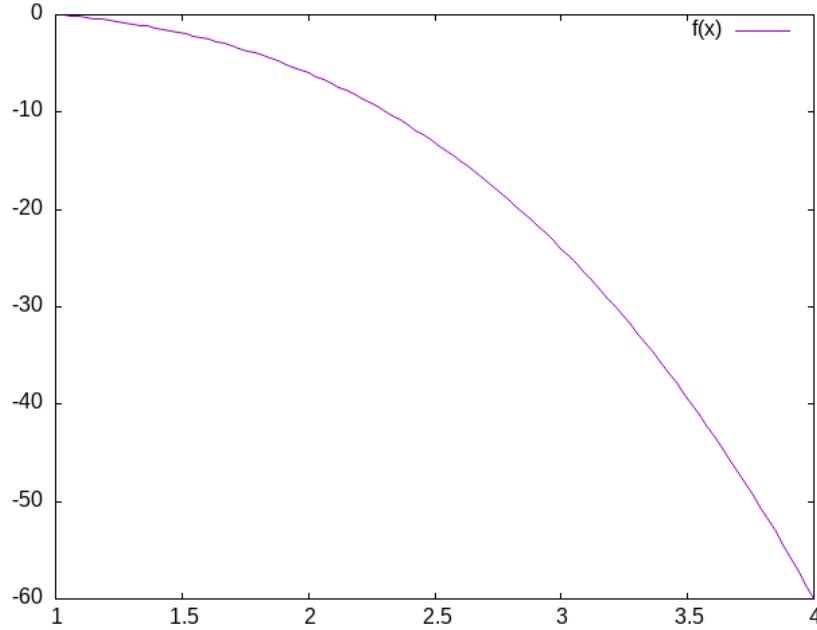


Figure 1: A plot of $f(x) = x - x^3$

What you want is some *function* of the variable x that will equal what you get when you take its derivative. For this function that is easy enough. You simply integrate. But what if we think of x as dynamic. It is not fixed. It varies with time, like the voltage inside a neuron, the air temperature, or the strength of a memory trace. Then your equation becomes, $\frac{dx}{dt} = x - x^3$. Integrate that.² In this case we can find an analytical³ solution. But we don't have to do much to our equations to make such solutions very difficult or impossible. For most of the use practical use cases in computational neuroscience we do not solve these equations analytically. Rather, we use the computer to approximate their behavior. Most of the equations in neuroscience, and biology generally for that matter, are well behaved enough that we can get away with this.

We will now review the action potential and lay the ground work for writing our own code to simulate a neuron's firing.

²It is not *that* hard, and Claude.AI (as of <2024-12-19 Thu>) can do it for you. But it is substantially harder than before.

³What does "analytical" mean here?

1.2 The Action Potential

Class Discussion

- 10 minutes to brush up on what an action potential is.
- Then be able to draw one.
- What are the axes?
- What ion causes the upward deflection?
- What causes the repolarization?
- Who discovered the action potential?
- Who won the Nobel Prize for characterizing the ionic events of the action potential experimentally and building a mathematical model?

Further Class Discussion

How is an action potential like a flush toilet?

1.3 A Mathematical Aside

I often find that psychology students let mathematical notation get in their way of understanding. The unfamiliar symbols lead people to skip the most important part of computational papers. The most important thing to learn is that mathematical notation is just like an emoji: a concise way to express a more complex idea. Once you learn the "slang" you can look beyond the characters. Here is an example of how simple math can start to look forbidding when notated.

What does $\sum_{x \in \{1,2,3\}} x = 6$ really mean?

Another confusing feature of mathematical notation is that there are often many ways to say the same thing and different mathematical communities have different conventions. But just because you cannot speak French you do not assume the French are really saying something different. So, when faced with a strange mathematical expression be open to the idea that it is just another way to say something you are already familiar with.

For example, $\frac{dx}{dt}$, $\dot{x}$, x' , $f'(x)$ are all different ways of talking about derivatives and derivatives are at the heart of differential equations.

1.4 Derivatives are Slopes

Class Discussion

1. What is a slope? 2. When in doubt return to definition. 3. Deriving the definition of a derivative. 4. What is the definition of a derivative?

1.4.1 Your computer is a tool for exploration

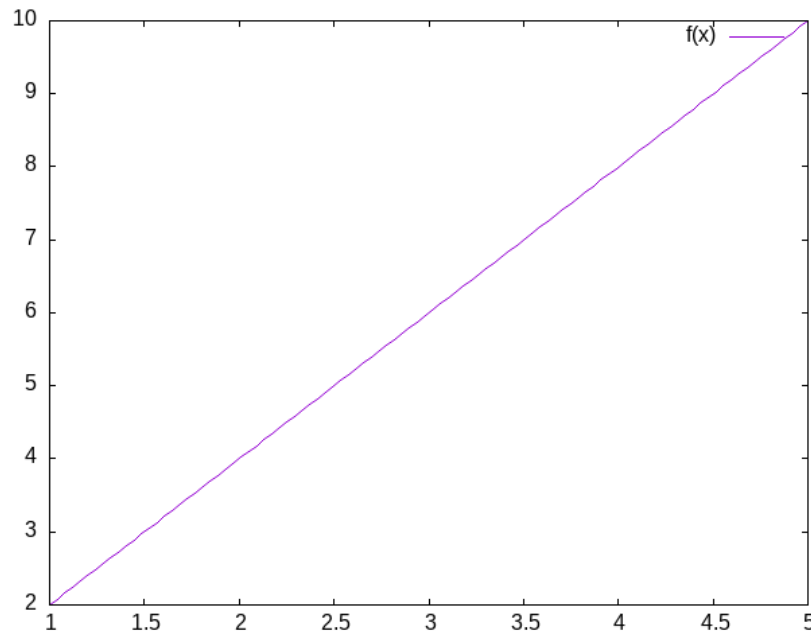


Figure 2: What is the slope of this line?

1.4.2 To approximate the derivative of a curve at a point draw a straight line close by

You pick two points that are "close enough" and you get an answer that is "close enough." If your answer isn't "close enough" then you move your points closer, until *in the limit* there is an infinitesimal distance between them.

Definition 1. *This is a test.*

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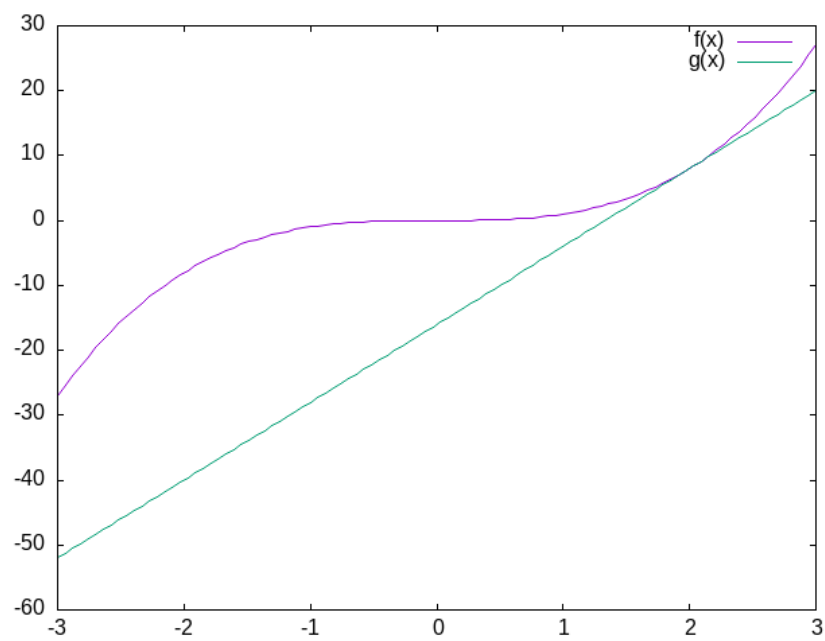


Figure 3: What is the slope of this curve? And what is missing from that question that you need to know to give an answer?